

YOONJU SIM

AI-assisted Optimization Researcher

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ABOUT ME

Yoonju Sim is an integrated M.S.-Ph.D. student in Industrial and Systems Engineering at KAIST and a member of the [Computational Optimization Methods \(COMET\) Lab](#) led by Prof. C Kwon. Her research focuses on designing AI-assisted algorithms for solving large-scale discrete optimization problems, particularly vehicle routing, but also broader MILP-based models. She uses techniques such as cutting planes and column generation together with AI, machine learning, and large language models (LLMs). Her goal is to build faster and more adaptive optimization methods by integrating classical OR principles with modern learning-based strategies.

RESEARCH INTERESTS

- Domain:**
- Transportation and Logistics
 - Quantum Computing
- Methods:**
- Mixed-Integer Programming based Approaches
 - Machine Learning
 - Large Language Models (LLMs)
- Programming:** Python, Julia, C

EDUCATION

2023.08 - Now	Korea Advanced Institute of Science and Technology (KAIST) Integrated M.S.-Ph.D. Student in Industrial and Systems Engineering	Daejeon, South Korea
2019.03 - 2023.08	Korea University (KU) Bachelor of Science in Industrial Management Engineering GPA: 4.4/4.5	Seoul, South Korea
2022.01 - 2022.05	Arizona State University Exchange student in Industrial Engineering	Arizona, US

RESEARCH EXPERIENCE

2023.08 - Now	Computational Optimization Methods (COMET) Lab, KAIST Advisor: Prof. C Kwon	Daejeon, South Korea
- Implementing large-scale optimization methods in Python - Investigating machine learning accelerated heuristics for vehicle routing problems - Executing projects to solve real-world business challenges		

WORK EXPERIENCE

2023.01 - 2022.07	Presales Intern @ SAP Korea Study spend management software, so-called 'Ariba' and assist in client meeting	Procurement Business Team
2022.07 - 2022.12	Presales Intern @ SAP Korea Study SAP Business Technology Platform and Low-code/No-code tools	Technology Team

PUBLICATIONS

- **Sim, Y.**, Berto, F., Hua, C., Park, J., and Kwon, C. *LLM-Driven Algorithm Design for Quantum Circuit Synthesis based on Binary Decision Diagrams*. **Transactions on Machine Learning Research (TMLR)**, 2026. [Paper](#) | [Code](#).

UNDER REVIEW

- **Sim, Y.**, Kim, H., and Kwon, C. *Test-Time Search for Neural Graph Coarsening Procedures for the CVRP*. **Under review**. [Preprint](#) | [Code](#).

WORKING PAPERS

- **Sim, Y.** et al. *A Computational Laboratory for AI-Driven Decomposition Design*. **In progress**.

PROJECT

2025.02 - 2025.09	Research on AI-accelerated Production Planning Developed an AI-accelerated Lagrangian relaxation model for CLSP, achieving near-optimal solutions and supporting fast 3-month production planning.	Samsung Electronics
2023.09 - 2023.12	Project on modeling UAM Simulation Designed and implemented a Python-based simulation to analyze UAM vertiport capacity, optimizing gate/FATO configurations using queueing and event-based modeling.	GS E&C
2023.01 - 2023.01	Research on a supplier performance management system Conduct a research on a supplier performance management system with respect to management consulting. Korea university, City of Vancouver, and UBC engaged Leader of a team of 6 students	IME Department, Korea University
2022.09 - 2022.12	Project about RUL prediction and Anomaly Detection Conduct a project to predict the residual life of equipment using unsupervised learning on sensor data. Win the first prize in the competition hosted by the IME Department, Korea University.	IME Department, Korea University
2021.06 - 2021.07	Project on Smart Campus - deploy a service named 'Subject Phd' Develop a website that assists students with course registration. Utilize text mining on syllabus data to make recommendation algorithm and elevate search algorithm. Win the first prize in the competition hosted by Korea University. Leader of a team of 4 students	Korea University

ACTIVITIES

2023.01 - 2023.12	Industrial Engineering Study Club in KU, WeTIE Serve as a vice president of the club with approximately 50 members and Organize sessions of introducing industrial engineering to the freshmen	Vice President
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CONFERENCE PRESENTATIONS

2025.10.22	Test-Time Search for Neural Graph Coarsening Procedures for the CVRP Oral Presentation	INFORMS Annual Meeting
2024.10.28	Charaterizing and Learning the Lagrangian Multipliers for the Bin Packing Problem Oral Presentation	INFORMS Annual Meeting

HONORS AND AWARDS

2025.11.18	Poster Competition for Industrial/Social Problem Solving Department of Industrial and Systems Engineering, KAIST	Excellence Award
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Last updated: September 5, 2026